SIMPUTER SIMPLIFIED

Will the brainchild of a small team of educationalists and technologists in Bangalore hold the solu to the Holy Grail of computing for the masses? We take apart the Simputer to see if it has what takes to bridge the Great Digital Divide

An amber sun casts its final rays over sleepy golden fields of mustard. A few wandering cows call out to their chocolate-coloured calves. Lambs skip through meadows under the watchful eye of their pigtailed mistress, the setting sun reflected in her eyes. Under the haystack, her father sits holding something that looks uncannily like a PDA, scribbling in some numbers with a stylus.

Engineers from the Indian Institute of Science (IISc), Bangalore, along with Encore Software Ltd conceptualised the Simputer (derived from the 'Simple Computer'), a Community Digital Assistant (CDA), to send technology to even the most inaccessible rural areas, where it could be used to perform applications as varied as village accounts management, crop budget maintenance and even education. Given the vast number of potential application areas, there are stringent demands placed on its design, construction and usability.

Quite like the PDA, it operates through touch, sound and simple visual icons, dispensing with the keyboard entirely. It incorporates support for Indian dialects and uses special speech processing technology to read the content out. The Simputer is made for multiple users, and individual settings can be retained by each user through the Smart Cards that the device supports. Using a non-proprietary operating system, software for specific applications can be developed for this device quickly.

Under the hood

To face the rigours of rural India and almost constant use, the Simputer uses some of the most proven technologies of mobile computing. It beats with the heart of a 32-bit Intel StrongARM SA-1100 processor, which features a clock speed of 200 MHz and is still low on power consumption. There is 32 MB of DRAM along with either 16 MB or 32 MB of external Flash RAM for permanent storage. It features an integrated 56 K V.90 modem and can be linked to a computer using its USB interface. Finally, it has a 320x240 resolution colour or monochrome LCD touch panel display.

While its specifications are like cal handheld's, it's interesting ho components have been handpicked t the highest levels of computing po the lowest of prices, while still maint a rich set of features. On the interface it packs speakers and microphone ja RJ-11 telephone jack, a Smart Card and even an IrDA wireless port. Keep mind the unreliability of our electrici device is powered by three AAA rechargeable batteries. The lack of ex storage is also taken care of by the n ous USB based external Flash device offer capacities ranging from 16 1 GB of storage.

Can I Buy One Yet?

Officially unveiled in April 2001, the Simputer is now ready to reach mass-production. Bharat Electronics Limited (BEL), one of the manufacturers of the device, is going to roll out 1,000 units to begin with and in due course, expand to 5,000 units per month. M.P. Mohammed, BEL's AGM (EM and D&E division) said BEL would manufacture the Simputer based on the specifications given by PicoPeta, which would be targeted at OEMs. PicoPeta is a consortium of technologists and the Simputer Trust, and plans to harness the potential of this device for the benefit of the society.

Besides developing system software bas on Linux under GPL (general public licer PicoPeta will also m the Simputer. Vinay Vishwanathan, Dire of PicoPeta, aims to mote the device as evolving platform for social change— tially we will target educational institu tions and e-governance projects in th country. Once the volumes pick up ar the prices come down from the prese 13,000 and Rs 15,000 for 16 MB and MB, respectively, we will look at retail sales." PicoPeta hopes to sell around 50,000 units this year.